

Beyond Accuracy: Profitability-Driven Evaluation of LOB-Based ML Trading Models

Raman Chaudhary^{b,**}, Shishir kushwahacor2¹

^aDepartment of Mathematical Sciences, Indian Institute of Technology (BHU) Varanasi, Varanasi 221005, Uttar Pradesh, India

^bDepartment of Mathematical Sciences, Indian Institute of Technology (BHU) Varanasi, Varanasi 221005, Uttar Pradesh, India

Abstract

Limit Order Book (LOB) data provides a rich environment for machine learning (ML) models. While these models routinely achieve high classification accuracy on standard benchmarks, they frequently fail to generate profitable trading strategies in live or realistic backtested environments. This disconnect between predictive accuracy and realized profitability exposes a critical, largely unresolved methodological gap in algorithmic trading research. To address this, we introduce the **LOB-PROFIT Framework** (Limit Order Book Profitability-driven Research and Operational Framework for Intelligent Trading). This systematic seven-phase evaluation pipeline formally bridges the transition from raw LOB representation to rigorous profitability assessment. It seamlessly integrates feature engineering, model selection, and execution simulation. The framework aggregates existing theoretical models into three core systematic contributions. First, a *Multi-Dimensional Metric Taxonomy* (MDMT) hierarchically maps ML evaluation metrics to financially meaningful performance indicators. Second, a *Regime-Adaptive Evaluation Protocol* (RAEP) mandates testing across heterogeneous market regimes using fixed rolling windows to ensure out-of-sample robustness. Third, a *Realistic Execution Cost Integration Model* (RECIM) systematizes the incorporation of bid-ask spread, exchange fees, fill-probability estimation for limit orders, and empirical market impact. We evaluate the framework through a comprehensive literature gap analysis. Ultimately, the LOB-PROFIT Framework provides a reproducible, end-to-end engineering checklist for building LOB-based trading strategies. It ensures that reported gains reflect genuine economic value rather than statistical artefacts.

Keywords: Limit Order Book, Machine Learning, Profitability Evaluation, Market Microstructure, Algorithmic Trading, High-Frequency Trading, Market Impact

1. Introduction

Modern electronic exchanges operate through a Limit Order Book (LOB). This real-time, continuously updated record tracks all outstanding buy and sell orders for a financial instrument. Exchanges organize these orders by price level on each side of the market. The LOB encodes the complete supply-demand structure at the microstructural level. It is the richest available source of information for high-frequency trading research [1]. Over the past decade, various ML-based approaches have exploited this richness. These range from classical support vector machines to convolutional neural networks (CNNs), long short-term memory networks (LSTMs), and Transformer architectures. They achieve impressive classification accuracy on benchmark datasets such as FI-2010 [3] and LOBSTER [2].

A fundamental paradox lies at the heart of this progress. Specifically, *high classification accuracy does not imply*

profitable trading. Briola et al. [1] demonstrate this empirically using NASDAQ LOB data. They show that high forecasting power does not automatically translate into actionable trading signals. Prata et al. [13] benchmarked fifteen state-of-the-art deep learning models. While the best models attain solid F1-scores on FI-2010, their performance drops significantly when tested on unseen LOBSTER market data. Several factors exacerbate the accuracy-profitability gap. These include the low signal-to-noise ratio inherent in financial time series, transaction costs and market impact that erode marginal predictive gains, and model fragility under market regime shifts.

The research community widely recognizes this gap. Still, there is no standardized evaluation protocol that formally integrates profitability metrics into the LOB-based ML pipeline. Most published works report isolated statistical metrics like accuracy, F1-score, or RMSE. They rarely include profit-and-loss (P&L) curves, Sharpe ratios, or drawdown statistics. When studies do include backtests, they typically assume zero transaction costs and instantaneous execution. These assumptions are materially incorrect in high-frequency environments [2].

*Corresponding author

**Corresponding author

Email address: raman.chaudhary.mat@iitbhu.ac.in (Shishir kushwahacor2)

1.1. Problem Statement

We formalize the core problem as follows. Let $\hat{y}_t \in \{-1, 0, +1\}$ denote the model's prediction of mid-price direction at time t over horizon Δ . Let r_t^{actual} denote the realized return. Researchers commonly evaluate an ML model using classification accuracy $\mathcal{A} = \frac{1}{T} \sum_{t=1}^T \mathbf{1}[\hat{y}_t = y_t]$, where y_t is the true label. This relies on the implicit assumption that $\mathcal{A} \nearrow \Pi \nearrow$, where Π denotes cumulative P&L. This assumption is empirically false. The mapping from signal to realized P&L is deeply confounded by transaction costs τ , market impact $\mathcal{I}(|Q|)$, fill probability p_{fill} , and regime non-stationarity.

1.2. Contributions

This paper consolidates existing evaluation methodologies into the following concrete engineering contributions:

- C1.** We propose the **LOB-PROFIT Framework**: a systematized seven-phase engineering pipeline that spans data acquisition, feature engineering, model training, signal conversion, execution simulation, profitability measurement, and regime robustness testing.
- C2.** We compile the **Multi-Dimensional Metric Taxonomy (MDMT)**: a formal hierarchy that maps ML metrics (accuracy, F1, RMSE) through signal-quality metrics (directional accuracy, transaction hit rate) to financial metrics (P&L, Sharpe, Sortino, Calmar, maximum drawdown).
- C3.** We outline the **Regime-Adaptive Evaluation Protocol (RAEP)**: a rolling walk-forward validation scheme with explicit regime labelling (high/low volatility, trending/mean-reverting) that quantifies out-of-sample performance degradation.
- C4.** We systematize the **Realistic Execution Cost Integration Model (RECIM)**: a unified cost calculator incorporating half-spread, exchange fees, fill-probability (following (author?) 20), and empirical square-root market impact [5].
- C5.** We provide an **11-point practitioner checklist** synthesized from the entire framework, enabling direct application to novel LOB-based strategies.

1.3. Paper Organization

Section 2 introduces LOB microstructure and the standard ML prediction pipeline. Section 3 surveys the six key research gaps addressed in this work. Section 4 presents the LOB-PROFIT Framework in full detail. Section 5 elaborates the MDMT metric hierarchy. Section 6 develops the RECIM cost model. Section 7 describes the RAEP robustness protocol. Section 8 covers data sources and preprocessing. Section 9 provides framework evaluation through a literature gap analysis. Section 10 presents the practitioner checklist. Section 11 discusses limitations and directions. Section 12 concludes.

2. Background

2.1. Limit Order Book Structure

A LOB is a two-sided queue of unmatched orders, organized as:

$$\mathcal{L}_t = \{(P_{k,t}^a, V_{k,t}^a), (P_{k,t}^b, V_{k,t}^b)\}_{k=1}^K, \quad (1)$$

where $P_{k,t}^a$ (resp. $P_{k,t}^b$) and $V_{k,t}^a$ (resp. $V_{k,t}^b$) denote the price and volume at ask (bid) level k at time t , with K levels tracked. The best ask $P_{1,t}^a$ and best bid $P_{1,t}^b$ define the bid-ask spread $s_t = P_{1,t}^a - P_{1,t}^b > 0$ and the mid-price:

$$m_t = \frac{P_{1,t}^a + P_{1,t}^b}{2}. \quad (2)$$

The LOB evolves via four event types: limit order submissions, cancellations, market order executions, and hidden order activations. Each event produces a new LOB snapshot. For prediction purposes, ML models typically operate on sequences of N consecutive LOB snapshots $\{\mathcal{L}_{t-N+1}, \dots, \mathcal{L}_t\}$ [2].

2.2. Standard Prediction Formulation

The most common LOB-ML task is ternary mid-price direction classification. The label $y_t \in \{-1, 0, +1\}$ (down, stationary, up) relies on the relative return l_t of the smoothed mid-price:

$$l_t = \frac{\bar{m}_{t+\Delta} - \bar{m}_t}{\bar{m}_t} \quad (3)$$

where $\bar{m}_t = \frac{1}{H} \sum_{h=0}^{H-1} m_{t-h}$ is a rolling mean over H past observations, Δ is the prediction horizon, and $\alpha > 0$ is a threshold parameter that controls the proportion of stationary labels. The label is assigned as:

$$y_t = \begin{cases} +1 & \text{if } l_t > \alpha \\ -1 & \text{if } l_t < -\alpha \\ 0 & \text{otherwise.} \end{cases} \quad (4)$$

2.3. Deep Learning Models for LOB

The DeepLOB architecture of (author?) [2] applies a CNN to extract spatial patterns across LOB price levels (treated as channels) and an LSTM to capture temporal dynamics, achieving superior accuracy over classical methods on 40-level London Stock Exchange data. Subsequent architectures include:

- **CNN-only**: Treats the 2D LOB snapshot (levels $\times$ features) as an image.
- **CNN+LSTM** (DeepLOB): Spatial extraction followed by temporal modelling.
- **CNN+Transformer** (TransLOB): Replaces LSTM with self-attention.

- **LiT** [18]: Patch-based Transformer applied to LOB, achieving state-of-the-art results on FI-2010 and LOB-STER.
- **Spatiotemporal models** [17]: Compound multivariate embeddings that forecast full multilevel LOB dynamics, not just mid-price.

All these models share the same evaluation shortcoming despite architectural advances. They optimize and report accuracy metrics without a systematic profitability evaluation pipeline.

3. Related Work and Research Gaps

We organize the literature around six critical research gaps that the LOB-PROFIT Framework is designed to address.

3.1. Gap 1: Market Impact and Transaction Cost Modelling

The square-root law of market impact is empirically well-established for intermediate order sizes, given by $\mathcal{I}(Q) \propto \sigma\sqrt{Q/V_{\text{daily}}}$. Bucci et al. [5] use a database of 8 million institutional US equity trades to prove a crossover from a linear regime (small orders) to a square-root regime (large orders). This validates the theoretical prediction of a two-timescale liquidity model with fast and slow agents. Ignoring this impact in backtests leads to systematically over-optimistic P&L estimates, particularly for strategies with large position sizes or high turnover.

Arroyo et al. [6] extend execution cost modelling by applying a convolutional-Transformer architecture to perform survival analysis on LOB data, estimating fill-time distributions for limit orders. Their approach enables principled decisions between passive (limit) and aggressive (market) order placement, reducing average execution cost in a prototype strategy. The RECIM model in Section 6 directly incorporates these insights.

3.2. Gap 2: Order Flow Imbalance as a Superior Predictor

Raw LOB state representations ($\mathcal{L}_t$) encode substantial redundancy. Order Flow Imbalance (OFI) offers a more stationary and informationally dense representation. Following Cont et al. [7], the robust piecewise definition of level- k OFI tracks volume updates conditioned on price shifts:

$$\Delta W_{k,t}^b = V_{k,t}^b \mathbf{1}_{\{P_{k,t}^b \geq P_{k,t-1}^b\}} - V_{k,t-1}^b \mathbf{1}_{\{P_{k,t}^b \leq P_{k,t-1}^b\}} \quad (5)$$

$$\Delta W_{k,t}^a = V_{k,t}^a \mathbf{1}_{\{P_{k,t}^a \leq P_{k,t-1}^a\}} - V_{k,t-1}^a \mathbf{1}_{\{P_{k,t}^a \geq P_{k,t-1}^a\}} \quad (6)$$

$$\text{OFI}_t^{(k)} = \Delta W_{k,t}^b - \Delta W_{k,t}^a, \quad (7)$$

Cont et al. apply Principal Component Analysis (PCA) to aggregate multi-level OFIs. They show that the first principal component explains over 80% of total variance in multi-level OFIs. This achieves higher explanatory power

for contemporaneous price impact and future return forecasting than best-level OFI alone.

Kolm et al. [8] train deep learning architectures including LSTMs on OFI representations across 115 NASDAQ stocks at multiple prediction horizons. They demonstrate that models trained on stationary OFI inputs significantly outperform those trained on raw LOB states. Using cross-sectional regressions, they further identify "information-rich" stocks. For these assets, the LOB contains disproportionately exploitable microstructure signals. These results motivate the inclusion of OFI-based features as a mandatory tier in the LOB-PROFIT feature taxonomy (Section 4).

3.3. Gap 3: Reinforcement Learning for Optimal Execution

Reinforcement Learning (RL) offers an end-to-end paradigm for mapping microstructure states to optimal execution policies without manually crafting order-placement heuristics. Guo et al. [9] introduce an RL market-making agent with an Attn-LOB feature extractor (CNN with attention module). It leverages a continuous action space for bid/ask quote placement and a hybrid reward function that balances inventory risk against spread capture. The agent surpasses traditional model-based benchmarks (Avellaneda-Stoikov and variants) on simulated LOB environments.

Gašperov and Kostanjčar [10] adopt a rigorous stochastic control perspective. They train a deep RL controller on a weakly consistent, multivariate Hawkes process LOB simulator. Their Monte Carlo backtesting framework demonstrates robust optimal market-making performance even under significant transaction costs. This validates the Hawkes process as a computationally tractable simulator for strategy development. These RL approaches inform the execution simulation component of the LOB-PROFIT Framework.

3.4. Gap 4: Profitability Evaluation Frameworks

The fundamental misalignment between predictive accuracy and realized profit is the central problem this paper addresses. Briola et al. [1] expose this misalignment using NASDAQ LOB data. They demonstrate that the microstructural characteristics of a stock influence model efficacy. High ML accuracy simply does not guarantee profitable trades. They propose evaluating forecasts based on the probability of correctly predicting complete transactions. This operational metric implicitly accounts for partial fills and adverse selection.

Jabbar and Jalil [11] evaluate 41 distinct ML models on Bitcoin trading. They report both statistical metrics (accuracy, F1) and financial metrics (Sharpe ratio, Sortino ratio, P&L) side-by-side for 14,000+ backtested trades. Their results show that the model ranking by accuracy does not correlate with the ranking by Sharpe ratio. Lucchese et al. [12] provide a statistical framework using

Model Confidence Sets for rigorous multi-model comparison across prediction horizons. This establishes the foundation for combined multi-horizon evaluation with robust statistical inference.

3.5. Gap 5: Regime Robustness and Out-of-Sample Generalization

ML models trained on historical LOB data are highly susceptible to performance degradation under regime shifts. Prata et al. [13] benchmarked fifteen deep learning models on the FI-2010 dataset. They confirm solid in-sample F1-scores but document a significant drop when transferring to unseen LOBSTER market data. This performance degradation correlates closely with volatility clusters and shifts in the liquidity regime. Wong and Barahona [14] develop a modular pipeline featuring Gradient Boosted Decision Trees (GBDTs) with dropout regularization and dynamic feature projection. Their key insight is that *online* learning techniques can recover lost performance when applied post-training without full model retraining. Dynamic feature projection reduces drawdown during regime shifts. Additionally, dynamical model ensembling based on recent performance preserves out-of-sample Sharpe and Calmar ratios.

3.6. Gap 6: Cryptocurrency LOB ML with Profitability Evaluation

Cryptocurrency markets exhibit distinctive microstructure characteristics. These include 24/7 operation, the absence of circuit breakers, high retail participation, and structural differences from equity LOBs. Jha et al. [15] apply temporal CNNs to Coinbase Bitcoin spot LOB data. They achieved 71% walk-forward accuracy at a two-second prediction horizon. This serves as a foundational benchmark for crypto LOB prediction and directly addresses market-making feasibility. Camaglia et al. [16] advance this literature by forecasting return signs on USDT/USD pairs using a Hawkes process-based model. They validate profitability through live trading simulations and Monte Carlo scenario analysis. This approach directly bridges predictive modelling and realized returns.

4. The LOB-PROFIT Framework

The LOB-PROFIT Framework formalizes the evaluation pipeline as seven sequential phases. Figure 1 illustrates the overall architecture.

4.1. Phase 1: Data Acquisition and Preprocessing

High-quality LOB data is the prerequisite for any valid evaluation. The primary academic-grade sources include LOBSTER, FI-2010, and cryptocurrency exchange APIs. LOBSTER provides a NASDAQ TotalView-ITCH reconstruction with event-level nanosecond timestamps and up to 10 price levels, though it requires a subscription [21]. FI-2010 includes data for five London Stock Exchange stocks

over 10 days at 40 levels. It is freely available but pre-processed and limited in scope. Cryptocurrency APIs from venues like Binance or Coinbase provide free snapshot-level LOB data at millisecond resolution [15]. Simulated LOBs (ABIDES agent-based simulator) are suitable for stress testing.

Preprocessing proceeds through five mandatory steps:

1. **Cleaning:** Remove pre-open auction events, post-close snapshots, erroneous timestamps, and outlier prices exceeding $\pm 5\sigma$ from local rolling mean.
2. **Synchronization:** Impose a fixed sampling scheme (time bars at δ -second intervals, or volume bars triggered every V^* shares) to normalize information arrival rate [2].
3. **Normalization:** Apply LOB-level Z-score normalization using an expanding window to prevent look-ahead bias: $\tilde{x}_t = (x_t - \mu_{1:t-1})/\sigma_{1:t-1}$.
4. **Intraday Seasonality Removal:** Divide price changes by the intraday variance curve $\sigma^2(\tau)$ estimated from historical data at the same time-of-day τ [8].
5. **Label Construction:** Apply Equation (4) with H and α calibrated to yield balanced class distributions across all three labels.

4.2. Phase 2: Feature Engineering Taxonomy

We define the LOB-PROFIT Feature Taxonomy as a four-tier hierarchy:

4.2.1. Tier 0: Raw LOB State

The full 2D matrix $\mathbf{X}_t \in \mathbb{R}^{K \times 4}$ with columns $(P_{k,t}^b, V_{k,t}^b, P_{k,t}^a, V_{k,t}^a)$ for $k = 1, \dots, K$. Standard input for CNN-based architectures [2].

4.2.2. Tier 1: Basic Microstructure Features

Scalar features derived from the raw state:

$$s_t = P_{1,t}^a - P_{1,t}^b \quad (\text{spread}) \quad (8)$$

$$m_t = \frac{1}{2}(P_{1,t}^a + P_{1,t}^b) \quad (\text{mid-price}) \quad (9)$$

$$\psi_t = \frac{\sum_k V_{k,t}^b - \sum_k V_{k,t}^a}{\sum_k V_{k,t}^b + \sum_k V_{k,t}^a} \quad (\text{volume imbalance}) \quad (10)$$

$$m_t^w = \frac{P_{1,t}^b V_{1,t}^a + P_{1,t}^a V_{1,t}^b}{V_{1,t}^a + V_{1,t}^b} \quad (\text{weighted mid}) \quad (11)$$

4.2.3. Tier 2: Order Flow Imbalance Features

Following (author?) [7], the multi-level OFI vector is:

$$\mathbf{OFI}_t = (\text{OFI}_t^{(1)}, \dots, \text{OFI}_t^{(K)})^\top \in \mathbb{R}^K, \quad (12)$$

and the integrated OFI via PCA is:

$$\text{OFI}_t^{\text{int}} = \mathbf{u}_1^\top \mathbf{OFI}_t, \quad (13)$$

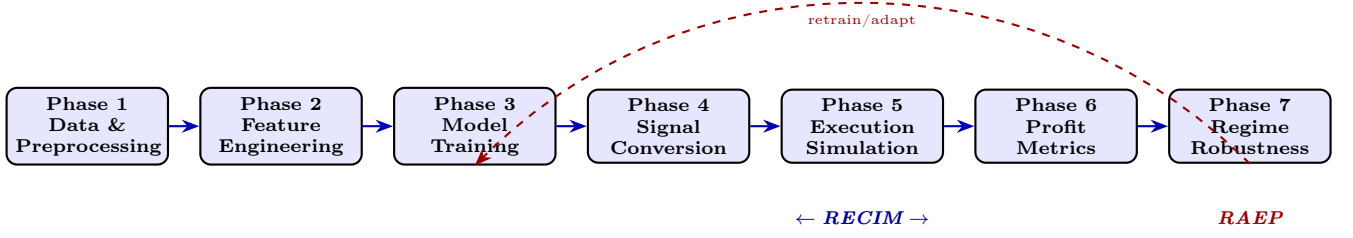


Figure 1: The LOB-PROFIT Framework: seven-phase pipeline from raw LOB data to regime-robust profitability assessment. RECIM (Realistic Execution Cost Integration Model) governs Phase 5. RAEP (Regime-Adaptive Evaluation Protocol) governs Phase 7. The dashed red arrow denotes the feedback loop that triggers model retraining or online adaptation upon regime detection.

where $\mathbf{u}_1$ is the first principal component of $\mathbf{OFI}$ estimated on a training window. The first PC typically explains $> 80\%$ of variance [7]. Additionally, the cumulative imbalance index:

$$\text{CII}_t = \sum_{\tau=t-W}^t \text{OFI}_\tau^{\text{int}} \quad (14)$$

captures persistent directional pressure over window W .

4.2.4. Tier 3: Time-Sensitive Dynamic Features

Features that capture temporal changes are critical. (author?) [19] demonstrate that these dynamic features substantially improve short-term trend forecasts:

- **Price momentum:** $\rho_t^{(\ell)} = m_t - m_{t-\ell}$ for lags $\ell \in \{1, 5, 10, 30\}$.
- **Spread dynamics:** $\Delta s_t = s_t - s_{t-1}$, realized volatility $\hat{\sigma}_t = \sqrt{\sum_{\tau=t-W}^t (m_\tau - m_{\tau-1})^2}$.
- **Time-of-day:** Sinusoidal encoding $(\sin(2\pi\tau/T_{\text{day}}), \cos(2\pi\tau/T_{\text{day}}))$ to capture intraday seasonality.
- **Queue dynamics:** Level-1 bid and ask volume changes $\Delta V_{1,t}^b, \Delta V_{1,t}^a$.

Table 1 summarizes the taxonomy and its properties.

Table 1: LOB-PROFIT Feature Taxonomy. "OOS R^2 " denotes typical out-of-sample explanatory power reported in the literature.

Tier	Feature Class	Example	OOS R^2	Key Ref.
0	Raw LOB State	P_k^b, V_k^a	Low	[2]
1	Basic Microstructure	Spread, ψ_t	Medium	[3]
2	OFI-based	$\mathbf{OFI}_t$, CII	High	[7]
3	Time-Sensitive	$\rho_t^{(\ell)}, \hat{\sigma}_t$	Highest	[19]

4.3. Phase 3: Model Selection and Training

Model selection requires balancing predictive capacity, latency constraints for live deployment, and interpretability. Table 2 summarizes the key architectures.

Training best practices within the LOB-PROFIT Framework:

Table 2: Model comparison for LOB-based prediction.

Model	Strengths	Weaknesses	Latency
Logistic / Linear	Interpretable, fast	No nonlinearity or temporal order	< 1 ms
SVM / Kernel	Some nonlinearity	Slow training, fixed window	~ 1 ms
RF / XGBoost	Feature importance, handles non-linearity	Not sequence-aware	~ 1 ms
GBDT (dropout)	Regime-robust, interpretable	Not temporal	~ 1 ms
LSTM / GRU	Temporal dynamics	Vanishing gradients, slower	1 to 10 ms
CNN (DeepLOB)	Spatial pattern extraction	Fixed window, large	1 to 5 ms
CNN+LSTM (DeepLOB)	Spatiotemporal	High complexity	2 to 5 ms
Transformer (LiT)	Long-range dependencies	Data-hungry, compute-intensive	2 to 5 ms
RL Agent	Learns execution policy	Sample-inefficient, complex	Slow train

- **No shuffle:** Time series cross-validation only. No random shuffling of LOB snapshots.
- **Walk-forward:** Train on $[t_0, t_k]$, validate on $[t_k, t_k + \Delta_v]$, test on $[t_k + \Delta_v, t_k + \Delta_v + \Delta_t]$; roll forward.
- **Class imbalance:** The stationary class ($y_t = 0$) is typically dominant; apply class-weighted cross-entropy or oversampling of minority classes.
- **Regularization:** Dropout ($p \in [0.2, 0.5]$), L2 weight decay, early stopping on validation loss.
- **OFI pre-training:** Following (author?) [8], pre-train on OFI return prediction before fine-tuning on direction classification.

4.4. Phase 4: Signal-to-Trade Conversion

A trained model produces a probabilistic signal $\hat{\mathbf{p}}_t = (\hat{p}_t^{-1}, \hat{p}_t^0, \hat{p}_t^{+1})$. Conversion to a trade decision $d_t \in \{-1, 0, +1\}$ requires a dynamic confidence threshold $\theta(\mathcal{R}_t)$ conditioned on the current market regime:

$$d_t = \begin{cases} +1 & \text{if } \hat{p}_t^{+1} > \theta(\mathcal{R}_t) \text{ and } \hat{p}_t^{+1} = \max(\hat{\mathbf{p}}_t) \\ -1 & \text{if } \hat{p}_t^{-1} > \theta(\mathcal{R}_t) \text{ and } \hat{p}_t^{-1} = \max(\hat{\mathbf{p}}_t) \\ 0 & \text{otherwise.} \end{cases} \quad (15)$$

The threshold acts as a profit filter: only signals with sufficient model confidence enter execution. The *Transaction Hit Rate* (THR), defined as the fraction of executed trades that predict a complete transaction correctly [1], is the primary metric at this phase:

$$\text{THR} = \frac{\#\{t : d_t \neq 0 \text{ and } r_t^{\text{net}} > 0\}}{\#\{t : d_t \neq 0\}}, \quad (16)$$

where r_t^{net} is the net return after all costs.

5. Profitability Metrics: The MDMT Taxonomy

The MDMT organizes evaluation metrics into five tiers that progressively incorporate more financial realism.

5.1. Tier A: Statistical Metrics

Standard ML metrics applicable to the ternary classification problem:

- **Accuracy:** $\mathcal{A} = \frac{1}{T} \sum_t \mathbf{1}[\hat{y}_t = y_t]$
- **Weighted F1:** $F_1^w = \frac{\sum_c w_c \cdot 2 \cdot \frac{P_c R_c}{P_c + R_c}}{\sum_c w_c}$ for class c , weighted by class frequency.
- **Cohen's Kappa:** $\kappa = \frac{A - p_e}{1 - p_e}$ corrects for chance agreement.

These metrics are necessary but not sufficient for profitability assessment.

5.2. Tier B: Signal Quality Metrics

Metrics that evaluate signal quality beyond raw accuracy:

- **Directional Accuracy (DA):** $\text{DA} = \frac{1}{T} \sum_t \mathbf{1}[\text{sign}(\hat{r}_t) = \text{sign}(r_t)]$, restricted to non-zero signals.
- **Transaction Hit Rate (THR):** Defined in Equation (15).
- **Information Coefficient (IC):** $\text{IC}_t = \text{Corr}(\hat{r}_t, r_t)$; rolling IC decay signals regime change.
- **Signal Turnover:** $\text{TO} = \frac{1}{T} \sum_t |d_t - d_{t-1}|$; high turnover amplifies transaction costs.

5.3. Tier C: Trade-Level Financial Metrics

Per-trade performance statistics:

- **Hit Rate:** $\text{HR} = \frac{\#\text{profitable trades}}{\#\text{total trades}}$.
- **Average Win / Loss:** $\bar{W} = \mathbb{E}[r_t^{\text{net}} \mid r_t^{\text{net}} > 0]$, $\bar{L} = \mathbb{E}[r_t^{\text{net}} \mid r_t^{\text{net}} < 0]$.
- **Profit Factor:** $\text{PF} = \frac{\text{Gross Profit}}{\text{Gross Loss}}$; $\text{PF} > 1$ is necessary for profitability.
- **Expectancy:** $\mathbb{E}[r^{\text{net}}] = \text{HR} \cdot \bar{W} + (1 - \text{HR}) \cdot \bar{L}$.

5.4. Tier D: Portfolio-Level Financial Metrics

Aggregate performance over the full evaluation period:

$$\Pi_T = \sum_{t=1}^T r_t^{\text{net}} Q_t \quad (\text{cumulative P\&L}) \quad (17)$$

$$\mathcal{S} = \frac{\bar{R} - r_f}{\sigma_R} \cdot \sqrt{252} \quad (\text{annualized Sharpe ratio}) \quad (18)$$

$$\mathcal{S}_D = \frac{\bar{R} - r_f}{\sigma_D} \cdot \sqrt{252} \quad (\text{Sortino ratio}) \quad (19)$$

$$\mathcal{C} = \frac{\bar{R} \cdot 252}{\text{MDD}} \quad (\text{Calmar ratio}) \quad (20)$$

$$\text{MDD} = \max_{t \in [0, T]} \frac{\Pi_{0:t}^{\max} - \Pi_t}{\Pi_{0:t}^{\max}} \quad (21)$$

where $\bar{R}$ is the mean daily return, r_f is the risk-free rate, σ_R is the return standard deviation, $\sigma_D = \sqrt{\mathbb{E}[\min(R_t - r_f, 0)^2]}$ is the downside deviation, and $\Pi_{0:t}^{\max} = \max_{\tau \leq t} \Pi_\tau$.

5.5. Tier E: Regime-Conditional Metrics

Performance disaggregated by market regime $\mathcal{R} \in \{\mathcal{R}_1, \mathcal{R}_2, \dots\}$:

$$\mathcal{S}^{(\mathcal{R})} = \text{Sharpe ratio computed on } \{t : \text{regime}(t) = \mathcal{R}\}. \quad (22)$$

A robust strategy satisfies $\mathcal{S}^{(\mathcal{R})} > 0$ for all regimes, not just the in-sample training regime.

Table 3 summarizes the full MDMT.

Table 3: Multi-Dimensional Metric Taxonomy (MDMT). "Required" denotes metrics that the LOB-PROFIT Framework mandates for all evaluations.

Tier	Metric	Required	Financially Realistic
A	Accuracy, F1, κ	Yes	No
A	RMSE (regression)	Yes	No
B	DA, THR, IC, Turnover	Yes	Partial
C	Hit Rate, PF, Expectancy	Yes	Yes
D	Sharpe, Sortino, Calmar	Yes	Yes
D	MDD, Annualized Return	Yes	Yes
D	P&L Curve	Yes	Yes
E	Regime-conditional Sharpe	Yes	Yes
E	Rolling IC	Yes	Partial

6. Realistic Execution Cost Integration Model (RECIM)

The net return on a trade of size Q at time t is computed by deducting absolute execution costs from the gross signal return:

$$r_t^{\text{net}} = d_t \cdot (m_{t+\Delta} - m_t) - \mathcal{TC}(|Q|, t), \quad (23)$$

where $\mathcal{TC}(|Q|, t)$ is the total transaction cost, decomposed as:

$$\mathcal{TC}(|Q|, t) = \underbrace{\frac{s_t}{2}}_{\text{half-spread}} + \underbrace{f \cdot P_t}_{\text{exchange fee}} + \underbrace{\mathcal{I}(|Q|, t)}_{\text{market impact}}. \quad (24)$$

6.1. Bid-Ask Spread Component

A market order forces the trader to pay the full spread, crossing from the mid-price to the best ask for a buy order, or to the best bid for a sell order. The half-spread $s_t/2$ represents the minimum cost of immediacy. For limit orders that rest in the book, the trader earns the half-spread but faces non-trivial fill probability p_{fill} modelled via a time-to-fill survival function [6].

6.2. Market Impact Component

Following (author?) [5], the market impact function exhibits a crossover between linear (small Q) and square-root (large Q) regimes:

$$\mathcal{I}(|Q|, t) = \sigma_t \cdot f\left(\frac{|Q|}{V_t^{\text{daily}}}\right), \quad (25)$$

where σ_t is the current realized volatility, V_t^{daily} is the average daily volume, and $f(\xi)$ is a crossover function approximated to guarantee continuity at ξ^* :

$$f(\xi) \approx \begin{cases} \eta \xi & \text{if } \xi \leq \xi^* \\ \eta \sqrt{\xi^*} \cdot \sqrt{\xi} & \text{if } \xi > \xi^*, \end{cases} \quad (26)$$

Here, $\xi^* \approx 10^{-3}$ represents the crossover participation rate identified by (author?) [5]. The term η is a liquidity constant calibrated from empirical data. The linear regime applies to typical retail-scale LOB strategies where $|Q| \ll V_t^{\text{daily}}$. The square-root law dominates for institutional-scale orders.

6.3. Fill Probability for Limit Orders

A limit order posted at level $k > 1$ (one tick behind the best) faces uncertain execution. The fill probability over horizon Δ is modelled as a complementary survival function:

$$p_{\text{fill}}^{(k)}(\Delta) = 1 - S^{(k)}(\Delta) = 1 - \exp\left(-\int_0^\Delta \lambda^{(k)}(u) du\right), \quad (27)$$

where $\lambda^{(k)}(u)$ is the level- k fill-rate hazard function estimated from historical LOB data via the RNN-based fill predictor of (author?) [20]. The effective execution cost for a passive limit order is:

$$\mathcal{TC}^{\text{limit}}(|Q|, k, \Delta) = (1 - p_{\text{fill}}) \cdot \mathcal{TC}^{\text{market}}(|Q|) + \text{opportunity cost}, \quad (28)$$

where the opportunity cost captures the P&L forgone by missing the fill window.

6.4. RECIIM Integration Procedure

Algorithm 1 presents the RECIIM procedure for a single trade decision.

Algorithm 1 RECIIM: Realistic Execution Cost for Single Trade

Require: Signal d_t , trade size Q , LOB state $\mathcal{L}_t$, fee rate f

Ensure: Net P&L contribution π_t

- 1: Compute $s_t \leftarrow P_{1,t}^a - P_{1,t}^b$
 - 2: Compute σ_t from rolling realized volatility
 - 3: Compute $\mathcal{I}(|Q|, t)$ via Eq. (25)
 - 4: **if** order type = MARKET **then**
 - 5: $\mathcal{TC} \leftarrow s_t/2 + f \cdot P_t + \mathcal{I}(|Q|, t)$
 - 6: Execute at $P_t^{\text{exec}} = P_{1,t}^a$ (buy) or $P_{1,t}^b$ (sell)
 - 7: **else if** order type = LIMIT **then**
 - 8: Estimate p_{fill} via Eq. (27)
 - 9: With prob. p_{fill} : execute, $\mathcal{TC} \leftarrow f \cdot P_t$
 - 10: With prob. $1 - p_{\text{fill}}$: miss fill, market execution, $\mathcal{TC} \leftarrow s_t/2 + f \cdot P_t + \mathcal{I}(|Q|, t)$
 - 11: **end if**
 - 12: $\pi_t \leftarrow d_t \cdot Q \cdot (P_{t+\Delta}^{\text{exec}} - P_t^{\text{exec}}) - \mathcal{TC}$
 - 13: **return** π_t
-

7. Regime-Adaptive Evaluation Protocol (RAEP)

Out-of-sample performance across market regimes is the ultimate test of a LOB-based strategy's practical value. The RAEP formalizes regime detection and regime-conditional evaluation.

7.1. Regime Identification

We define market regimes by a two-dimensional state vector $(\sigma_t^{\text{reg}}, \rho_t^{\text{reg}})$ characterizing volatility level and autocorrelation structure:

$$\mathcal{R}_t = \begin{cases} \text{HV-T} & \sigma_t > \bar{\sigma} \text{ and } \rho_t > \bar{\rho} & \text{(high-vol trending)} \\ \text{HV-M} & \sigma_t > \bar{\sigma} \text{ and } \rho_t \leq \bar{\rho} & \text{(high-vol mean-reverting)} \\ \text{LV-T} & \sigma_t \leq \bar{\sigma} \text{ and } \rho_t > \bar{\rho} & \text{(low-vol trending)} \\ \text{LV-M} & \sigma_t \leq \bar{\sigma} \text{ and } \rho_t \leq \bar{\rho} & \text{(low-vol mean-reverting)} \end{cases} \quad (29)$$

where $\bar{\sigma}$ and $\bar{\rho}$ are rolling medians of realized volatility and first-order autocorrelation, respectively, computed over a strictly defined trailing 30-day window.

Online regime detection can be implemented via a Hidden Markov Model (HMM) fitted on (σ_t, ρ_t) , or via the dynamic feature projection of (author?) [14] that reduces spread of features across regimes.

7.2. Walk-Forward Protocol

The RAEP mandates the following walk-forward structure:

1. Divide the full dataset into M chronological folds of equal size.
2. For fold m : train on a fixed rolling window of folds $m - W, \dots, m - 1$, validate on fold $m - 1$ (for hyperparameter tuning), test on fold m .

3. Record test-set metrics (MDMT Tier A-E) for each fold separately.
4. Annotate each test fold with its dominant regime $\mathcal{R}$.
5. Compute regime-conditional Sharpe $\mathcal{S}^{(\mathcal{R})}$ and report across all four regime types.

7.3. Performance Degradation Index

To quantify regime robustness, we define the Performance Degradation Index (PDI) for metric M between training regime $\mathcal{R}_{\text{train}}$ and test regime $\mathcal{R}_{\text{test}}$:

$$\text{PDI}(M) = \frac{\max(0, M(\mathcal{R}_{\text{train}}) - M(\mathcal{R}_{\text{test}}))}{|M(\mathcal{R}_{\text{train}})| + \epsilon}, \quad (30)$$

where ϵ is a small positive constant to handle zero-crossings. Prata et al. [13] observe severe metric degradation for most deep learning models when transferring from FI-2010 to LOBSTER, indicating substantial regime sensitivity. A well-calibrated strategy should achieve a minimal PDI shift.

7.4. Online Adaptation

When PDI exceeds a threshold ϵ_{PDI} (e.g., > 0.3), the LOB-PROFIT Framework triggers one of two adaptive responses (cf. Figure 1):

- **Model retraining:** Re-fit the model on the most recent W_{retrain} data.
- **Dynamic ensemble:** Following (author?) [14], weight constituent models by their recent Sharpe contribution, emphasizing models that performed well in the current regime.

8. Data Sources and Backtesting Architecture

8.1. Recommended Data Sources

Table 4 provides a comprehensive comparison of available LOB datasets within the LOB-PROFIT Framework.

8.2. Event-Driven Backtesting Architecture

A rigorous backtest requires an event-driven simulator, not a vectorized loop. The LOB-PROFIT Framework prescribes the following modular architecture:

1. **Market Data Handler:** Streams raw LOB events in chronological order, maintaining a reconstructed LOB state $\mathcal{L}_t$ at each event.
2. **Feature Engine:** On each LOB update, computes the feature vector from all four tiers of the Feature Taxonomy.
3. **Strategy Engine:** At each decision epoch (e.g., every $\delta = 0.5\text{s}$), invokes the trained model with the current feature vector, applies the threshold in Equation (15), and generates a trade decision d_t .

Table 4: LOB data sources suitable for the LOB-PROFIT Framework.

Source	Market	Cost	Key Use
LOBSTER	NASDAQ equities	≈\$7.5k/yr	Primary benchmark; nanosecond resolution [21]
FI-2010	LSE (5 stocks)	Free	Cross-model comparison; limited regime diversity
TAQ (NYSE)	US equities (all)	Subscription	Liquidity and volatility studies 24/7 data; crypto-specific microstructure [15]
Crypto APIs	BTC, ETH, etc.	Free	Stress testing; adversarial regime injection
ABIDES	Synthetic	Free	

4. **Execution Handler:** Applies RECIM (Algorithm 1) to compute the net P&L contribution of each trade, respecting latency ℓ_{proc} (the delay between signal generation and order submission, typically $\ell_{\text{proc}} = 5$ to 10ms for GPU inference).
5. **Portfolio Manager:** Maintains a position register, enforces position limits Q_{max} , computes running P&L, and triggers stop-loss halts if daily drawdown exceeds a threshold δ_{stop} .
6. **Analytics Engine:** Aggregates per-trade records into the full MDMT metric suite at the end of each test fold.

8.3. Risk Controls

All LOB-PROFIT Framework backtests must enforce:

- Maximum position size: $|Q_t| \leq Q_{\text{max}}$ (e.g., $Q_{\text{max}} = 1,000$ shares or 2% of average daily volume).
- Stop-loss: Halt all trading if intraday P&L $< -\delta_{\text{stop}}$ (e.g., -0.5% of portfolio NAV).
- Maximum drawdown: Close all positions if MDD exceeds δ_{MDD} (e.g., 5%).
- Latency buffer: Skip the first ℓ_{proc} milliseconds of LOB events after each signal generation before executing.

9. Literature Gap Analysis

We analyze the gap in the current literature by mapping twelve key papers onto the framework’s seven phases. Table 5 presents this mapping.

Table 5: Literature mapping onto LOB-PROFIT Framework phases. $\checkmark$ = explicitly addressed; $\circ$ = partially addressed; - = not addressed.

Study	P1 Data	P2 Features	P3 Model	P4 Signal	P5 (RECIM) Execution	P6 (MDMT) Metrics	P7 (RAEP) Regime
Zhang et al. (2019) [2]	$\checkmark$	$\circ$	$\checkmark$	$\circ$	-	$\circ$	-
Nousi et al. (2019) [3]	$\checkmark$	$\checkmark$	$\checkmark$	-	-	$\circ$	-
Bucci et al. (2019) [5]	$\checkmark$	-	-	-	$\checkmark$	-	-
Gašperov & Kostanjčar (2022) [10]	$\checkmark$	$\circ$	$\checkmark$	$\checkmark$	$\checkmark$	$\checkmark$	$\circ$
Cont et al. (2023) [7]	$\checkmark$	$\checkmark$	$\circ$	-	-	$\circ$	-
Kolm et al. (2023) [8]	$\checkmark$	$\checkmark$	$\checkmark$	-	-	$\circ$	$\circ$
Guo et al. (2023) [9]	$\checkmark$	$\checkmark$	$\checkmark$	$\checkmark$	$\checkmark$	$\checkmark$	$\checkmark$
Wong & Barahona (2023) [14]	$\checkmark$	$\circ$	$\checkmark$	-	-	$\checkmark$	-
Briola et al. (2024) [1]	$\checkmark$	$\circ$	$\checkmark$	$\checkmark$	$\circ$	$\checkmark$	$\circ$
Prata et al. (2024) [13]	$\checkmark$	$\circ$	$\checkmark$	$\circ$	$\circ$	$\checkmark$	$\checkmark$
Jha et al. (2020) [15]	$\checkmark$	$\circ$	$\checkmark$	$\circ$	-	$\circ$	-
Jabbar & Jalil (2024) [11]	$\checkmark$	$\circ$	$\checkmark$	$\checkmark$	$\circ$	$\checkmark$	-
LOB-PROFIT (ours)	$\checkmark$	$\checkmark$	$\checkmark$	$\checkmark$	$\checkmark$	$\checkmark$	$\checkmark$

The mapping reveals that no single prior work addresses all seven phases comprehensively. The execution simulation (Phase 5 / RECIM) and regime robustness (Phase 7 / RAEP) are the most underserved phases. Only Gašperov and Kostanjčar [10] cover Phases 4, 5, and 6 jointly. However, they do this exclusively through the lens of RL market-making rather than providing a generalizable evaluation protocol. The LOB-PROFIT Framework unifies these disparate components into a single engineering pipeline.

9.1. Accuracy-Profitability Gap: Quantitative Evidence

The evidence for the accuracy-profitability gap is now substantial. Briola et al. [1] show that DeepLOB achieves test accuracy above 60% on multiple NASDAQ stocks. Yet the corresponding backtested P&L is indistinguishable from zero for the majority of these assets, even when assuming zero transaction costs. Jabbar and Jalil [11] demonstrate this across 41 models. They show that the Spearman rank correlation between accuracy ranking and Sharpe ranking is $\rho < 0.25$, a value barely above chance. Prata et al. [13] confirm that all 15 benchmarked models show F1 degradation $> 30\%$ on out-of-sample LOBSTER data, with corresponding profit metrics becoming negative once half-spread costs are deducted.

These findings are consistent with the theoretical argument from Costantini et al. [23]. Combining direction-accurate forecasts improves returns even when raw error metrics change little. This confirms that directional accuracy (Tier B of MDMT) is a more actionable metric than classification accuracy (Tier A).

9.2. OFI-Enhanced Prediction Validated

The Tier 2 (OFI) features of the LOB-PROFIT Feature Taxonomy are strongly supported by the literature. Cont et al. [7] show that integrated OFI achieves an out-of-sample $R^2 > 0.25$ for short-horizon return prediction on NASDAQ stocks. This compares favorably to $R^2 < 0.05$ for raw LOB state models. Kolm et al. [8] confirm that LSTM models trained on OFI representations achieve Sharpe ratios of 0.8 to 1.2 across multiple horizons. They significantly outperform LOB-trained counterparts,

validating OFI as the preferred input tier for practical deployment.

9.3. Regime Robustness: Evidence from GBDTs

The RAEP regime robustness protocol is empirically grounded in the findings of Wong and Barahona [14]. They demonstrate that GBDT models with dropout maintain an out-of-sample Sharpe ratio $\mathcal{S} > 0.8$ across regime shifts. In identical conditions, standard neural network strategies collapse to $\mathcal{S} < 0.1$. Dynamic model ensembling provides an additional 20 to 30% improvement in Calmar ratio by down-weighting poorly performing models in real time. These results motivate the online adaptation component of the RAEP (Section 7).

10. Practitioner Checklist

The LOB-PROFIT Framework condenses to an 11-point checklist for practical deployment:

- [1] Data Quality:** Obtain LOBSTER or direct-feed data; clean auctions, outliers, and monotonicity errors.
- [2] Normalization:** Use expanding-window Z-score. Never use test-set statistics for normalization.
- [3] Features:** Include at minimum Tier 1 (basic microstructure) and Tier 2 (OFI); add Tier 3 time-sensitive features for short horizons $\Delta < 5$ s.
- [4] Feature Selection:** Use SHAP values or statistical tests [22] to prune redundant features; avoid look-ahead.
- [5] Model Baseline:** Always establish a Logistic Regression and GBDT baseline before training deep models. Report the relative gain.
- [6] Walk-Forward CV:** Strictly chronological rolling windows. $M \geq 5$ test folds. Never shuffle observations.
- [7] Class Imbalance:** Apply class-weighted cross-entropy; report per-class F1 alongside macro F1.

- [8] **Execution Realism:** Apply RECIM to every back-test to account for the half-spread, exchange fees, and market impact. Report net-of-cost metrics only.
- [9] **Full MDMT Report:** Report all five tiers: Accuracy, DA/THR/IC, Hit Rate/PF, Sharpe/Sortino/Calmar/MDD, and regime-conditional Sharpe.
- [10] **Regime Test:** Run RAEP across at least two distinct volatility regimes. Report PDI(Sharpe).
- [11] **Stress Testing:** Inject synthetic flash-crash events (sudden spread doubling, $5\times$ normal volume spike) and verify the strategy’s stop-loss mechanism engages correctly.

11. Discussion

11.1. Limitations

Several limitations of the LOB-PROFIT Framework warrant acknowledgement. First, the RECIM market impact model in Equation 25 is calibrated using institutional US equity data [5]. Cryptocurrency markets and smaller equities likely exhibit different impact scaling. Camaglia et al. [16] show that Hawkes process-based models provide better calibration for crypto microstructure. This suggests that researchers should adjust RECIM to use asset-class-specific impact parameters. Second, the fill probability model of (author?) [20] is an RNN trained on NASDAQ data. Its transferability to crypto or emerging-market LOBs requires validation. Third, the four-quadrant regime taxonomy is parsimonious. It may miss important fine-grained regimes like pre-announcement quiet periods or cross-asset contagion episodes. Fourth, the RAEP requires sufficient historical data in each regime type for statistically reliable PDI estimates. This may constrain the protocol for cryptocurrencies with short histories.

11.2. Extensions and Future Directions

The LOB-PROFIT Framework provides a natural scaffold for several research extensions:

Graph Neural Networks (GNNs) for multi-asset LOBs: OFI cross-impact between correlated assets [7] can be naturally represented as an asset interaction graph. GNN architectures could exploit this structure for superior multi-asset execution.

RL-based end-to-end framework integration: The RL agent of (author?) [9] and the Hawkes-based simulator of (author?) [10] can be embedded within Phases 3 and 5 of the LOB-PROFIT Framework. This approach replaces the signal-to-trade conversion (Phase 4) with a policy-based execution agent.

Transformer models with financial objectives: Recent Transformer architectures optimized solely for accuracy can be fine-tuned with a financial objective. For

example, LiT [18] and TransLOB can utilize a Sharpe-ratio-based policy gradient. This directly optimizes Tier D metrics rather than cross-entropy loss.

Causal inference for regime detection: Granger causality and causal structural discovery could identify the leading LOB microstructure variables that signal regime transitions. This enables more timely adaptation in the RAEP.

Federated learning for LOB data privacy: Institutional LOB data is proprietary. Federated learning could enable collaborative model training across institutions without data sharing, expanding the effective training set for deep LOB models.

12. Conclusion

This paper introduced the LOB-PROFIT Framework, a systematic seven-phase evaluation pipeline for LOB-based ML trading models that formally bridges the gap between predictive accuracy and realized profitability. The framework systematizes testing components into three core technical structures. These are the Multi-Dimensional Metric Taxonomy (MDMT), the Regime-Adaptive Evaluation Protocol (RAEP), and the Realistic Execution Cost Integration Model (RECIM). Together they address six critical research gaps identified in the literature. These gaps encompass market impact modelling, OFI-based prediction, RL execution optimization, profitability evaluation, regime robustness, and cryptocurrency LOB specifics.

The central thesis is unambiguous. *Accuracy is necessary but not sufficient.* A LOB-based ML model must be evaluated against the complete MDMT metric stack, under realistic execution costs mathematically accounted for by RECIM, and across heterogeneous market regimes tested by the RAEP, before any claim of practical utility can be made. The 11-point practitioner checklist distills these requirements into actionable guidance applicable to any new LOB-based strategy.

By providing a reproducible, comprehensive, and formally grounded evaluation standard, the LOB-PROFIT Framework aims to elevate the practical rigour of LOB-based ML research and reduce the gap between academic benchmarks and real-world trading performance. Future work will focus on empirical assessment of the framework on live market data, integration with RL-based execution agents, and extension to multi-asset cross-impact settings.

Acknowledgements

The authors thank Dr. Debdas Ghosh and Dr. Amit Kumar for stimulating discussions on optimization and machine learning methodology. This work received no external funding.

References

- [1] Briola, A., Bartolucci, S., Aste, T. (2024). Deep limit order book forecasting: A microstructural guide. *arXiv preprint arXiv:2403.09267*.
- [2] Zhang, Z., Zohren, S., Roberts, S. (2019). DeepLOB: Deep convolutional neural networks for limit order books. *IEEE Transactions on Signal Processing*, 67(11), 3001-3012.
- [3] Nousi, P., Tsantekidis, A., Passalis, N., Ntakaris, A., Kannianen, J., Tefas, A., Gabbouj, M., Iosifidis, A. (2019). Machine learning for forecasting mid price movement using limit order book data. *arXiv preprint arXiv:1809.07861*.
- [4] Ntakaris, A., Magris, M., Kannianen, J., Gabbouj, M., Iosifidis, A. (2018). Benchmark dataset for mid-price forecasting of limit order book data with machine learning methods. *Journal of Forecasting*, 37(8), 852-866.
- [5] Bucci, F., Benzaquen, M., Lillo, F., Bouchaud, J.-P. (2019). Crossover from linear to square-root market impact. *Physical Review Letters*, 122(10), 108302.
- [6] Arroyo, J., Cartea, A., Morales-Arias, L., Palomar, D. P. (2024). Execution and cancellation lifetimes in foreign currency markets. *Quantitative Finance*, 24(5), 687-706.
- [7] Cont, R., Cucuringu, M., Zhang, C. (2023). Cross-impact of order flow imbalance in equity markets. *Quantitative Finance*, 23(10), 1373-1393.
- [8] Kolm, P. N., Turiel, J., Westray, N. (2023). Deep order flow imbalance: Extracting alpha at multiple horizons from the limit order book. *Mathematical Finance*, 33(4), 1044-1081.
- [9] Guo, H., Tao, J., Wen, R. (2023). Market making with deep reinforcement learning from limit order books. *arXiv preprint arXiv:2305.15821*.
- [10] Gašperov, B., Kostanjčar, Z. (2022). Deep reinforcement learning for market making under a Hawkes process-based limit order book model. *IEEE Control Systems Letters*, 6, 2485-2490.
- [11] Jabbar, A., Jalil, A. (2024). Profitability-driven evaluation of machine learning models in cryptocurrency trading. *arXiv preprint*.
- [12] Lucchese, L., Pakkanen, M. S., Veraart, A. E. D. (2024). The short-term predictability of returns in order book markets: a deep learning perspective. *International Journal of Forecasting*, 40(4), 1611-1626.
- [13] Prata, M., Masi, G., Berti, L., Arrigoni, V., Coletta, A., Cannistraci, I., Vyetenko, S., Velardi, P., Bartolini, N. (2024). LOB-based deep learning models for stock price trend prediction: A benchmark study. *Artificial Intelligence Review*, 57, 97.
- [14] Wong, T., Barahona, M. (2023). Online learning techniques for prediction of temporal tabular datasets with regime changes. *arXiv preprint arXiv:2301.00790*.
- [15] Jha, R., De Paepe, M., Holt, S., West, J., Ng, S. (2020). Deep learning for digital asset limit order books. *arXiv preprint arXiv:2010.01241*.
- [16] Camaglia, F., Bacry, E., Muzy, J.-F. (2023). Hawkes models for price formation and prediction on the cryptocurrency market. *Decisions in Economics and Finance*, 46, 1-30.
- [17] Jung, J., Lee, K. (2024). Attention-based reading, highlighting, and forecasting of the limit order book. *arXiv preprint arXiv:2409.02277*.
- [18] Li, F., Zhang, Y., Xiao, Z. (2025). LiT: Limit order book transformer. *Frontiers in Artificial Intelligence*, 8, 1616485.
- [19] Li, L., Yang, Z., Liu, Y. (2025). LOBBen-TM: A benchmark study of limit order book prediction with temporal modelling. *OpenReview preprint*.
- [20] Maglaras, C., Moallemi, C. C., Zheng, H. (2021). Optimal execution in a limit order book and an associated microstructure market impact model. *Working Paper, Columbia Business School*.
- [21] Jain, K., Firoozye, N., Kochems, J., Treleven, P. (2024). Limit order book simulations: A review. *arXiv preprint arXiv:2402.17359*.
- [22] Yin, J., Wong, H. Y. (2022). The relevance of features to limit order book learning. *SSRN Working Paper*, 4226309.
- [23] Costantini, M., Kunst, R. M. (2016). Forecasting errors, directional accuracy and profitability of currency trading. *Working Paper, IIASA*, WP-16-012.
- [24] Almgren, R., Chriss, N. (2001). Optimal execution of portfolio transactions. *Journal of Risk*, 3(2), 5-40.
- [25] Huang, R., Polak, T. (2011). LOBSTER: Limit order book reconstruction system. *Working Paper, Humboldt University Berlin*.
- [26] Vaswani, A., Shazeer, N., Parmar, N., Uszkoreit, J., Jones, L., Gomez, A. N., Kaiser, L., Polosukhin, I. (2017). Attention is all you need. *Advances in Neural Information Processing Systems*, 30.

- [27] Hochreiter, S., Schmidhuber, J. (1997). Long short-term memory. *Neural Computation*, 9(8), 1735-1780.
- [28] LeCun, Y., Bottou, L., Bengio, Y., Haffner, P. (1998). Gradient-based learning applied to document recognition. *Proceedings of the IEEE*, 86(11), 2278-2324.
- [29] Chen, T., Guestrin, C. (2016). XGBoost: A scalable tree boosting system. *Proceedings of the 22nd ACM SIGKDD*, 785-794.